

# Independent Non-Redundant Open Conjectures in Number Theory

## A Minimal Core of 115 Pairwise Independent Problems

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<sup>1</sup>*Independent Compilation*  
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We list 115 open conjectures in number theory that are pairwise independent in the sense that no one is known to imply another after collapsing standard implication chains: Grand RH  $\Rightarrow$  RH, GRH, ERH, Lindelöf, Density; Firoozbakht  $\Rightarrow$  Cramér, Legendre, Oppermann, Andrica, Brocard-gap; Bateman-Horn  $\Rightarrow$  Schinzel H  $\Rightarrow$  Dickson  $\Rightarrow$  Bunyakovsky  $\Rightarrow$   $k$ -tuple  $\Rightarrow$  Twin, Polignac  $k$ ,  $n^2 + 1$ ; Vojta  $\Rightarrow$  abc  $\Leftrightarrow$  Szpiro  $\Rightarrow$  Pillai, Hall, Fermat-Catalan; Bloch-Kato  $\Rightarrow$  BSD  $\Rightarrow$  Parity, Sha finiteness; Standard  $\Rightarrow$  Hodge, Tate; Sarnak  $\Rightarrow$  Chowla; Schanuel  $\Rightarrow$  Four Exponentials,  $e + \pi$ ,  $e^e$  transcendence. To maximize independent count we drop the 5 very general polynomial-prime conjectures and keep specific prime families, and keep the strongest member of each remaining chain. Weak Goldbach (Helfgott 2013) is excluded as resolved.

### I. IMPLICATION COLLAPSE

- RH family kept as Grand RH.
- Prime gaps family kept as Firoozbakht.
- Polynomial primes family dropped (Bateman-Horn, Schinzel H, Dickson, Bunyakovsky,  $k$ -tuple) to keep specific forms independent.
- Polignac kept, Twin/Cousin/Sexy dropped.
- abc family kept as Vojta.
- BSD family kept as Bloch-Kato.
- Cycles family kept as Standard Conjectures.
- Möbius family kept as Sarnak.
- Transcendence family kept as Schanuel.

### II. INDEPENDENT LIST

1. **Grand Riemann Hypothesis:** Every non-trivial zero of any automorphic  $L$ -function  $L(s, \pi)$  has  $\Re(s) = 1/2$ .
2. **Firoozbakht's Conjecture:**  $p_n^{1/n}$  strictly decreasing:  $p_{n+1}^{1/(n+1)} < p_n^{1/n}$  for  $n \geq 1$ .
3. **Fortune's Conjecture:** Let  $p_n\# = 2 \cdot 3 \cdots p_n$ . Let  $m > 1$  minimal with  $p_n\# + m$  prime. Then  $m$  (Fortunate number) is prime.
4. **Gilbreath's Conjecture:** Starting from primes  $a_0(n) = p_n$ , define  $a_{k+1}(n) = |a_k(n+1) - a_k(n)|$ . Then  $a_k(1) = 1$  for all  $k \geq 1$ .
5. **Grimm's Conjecture:** If  $n+1, \dots, n+k$  are composite, there exist distinct primes  $p_i$  with  $p_i \mid n+i$ .
6. **Artin Primitive Root:** For  $a \neq -1$  and not a perfect square,  $a$  is primitive root mod  $p$  for infinitely many  $p$ .
7. **Mersenne infinitude:** Infinitely many primes  $M_p = 2^p - 1$ ,  $p$  prime.
8. **Fermat finiteness:** Only five Fermat primes  $F_n = 2^{2^n} + 1$  for  $n = 0, \dots, 4$ : 3, 5, 17, 257, 65537.
9. **Repunit prime infinitude:** Infinitely many primes  $R_n = (10^n - 1)/9$  base 10.
10. **Wagstaff prime infinitude:** Infinitely many primes  $(2^p + 1)/3$ ,  $p$  prime.
11. **Fibonacci prime infinitude:** Infinitely many  $F_n$  prime,  $F_1 = F_2 = 1$ ,  $F_{n+2} = F_{n+1} + F_n$ .
12. **Cullen prime infinitude:** Infinitely many primes  $C_n = n2^n + 1$ .
13. **Woodall prime infinitude:** Infinitely many primes  $W_n = n2^n - 1$ .
14. **Wilson prime infinitude:**  $p^2 \mid (p-1)! + 1$ . Only 5, 13, 563 known. Infinitely many?
15. **Wolstenholme prime infinitude:**  $p$  with  $\binom{2p-1}{p-1} \equiv 1 \pmod{p^4}$ . Only 16843, 2124679 known.
16. **Wall-Sun-Sun existence:** Does prime  $p$  exist with  $p^2 \mid F_{p-(\frac{p}{5})}$  where  $F$  Fibonacci and  $(\frac{p}{5})$  Legendre symbol?
17. **Wieferich prime infinitude:**  $2^{p-1} \equiv 1 \pmod{p^2}$ . Infinitely many.
18. **Palindromic prime infinitude base 10:** Prime with decimal expansion reverse equal itself, infinitely many.
19. **Emirp infinitude:** Prime  $p$  with  $\text{reverse}_{10}(p) \neq p$  prime, infinitely many.
20. **Pierpont prime infinitude:** Primes  $2^u 3^v + 1$  infinitely many.
21. **Carol/Kynea infinitude:** Carol  $(2^n - 1)^2 - 2$  and Kynea  $(2^n + 1)^2 - 2$  prime infinitely often.
22. **Factorial prime infinitude:**  $n! \pm 1$  prime infinitely often.

- 23. Primorial prime infinitude:**  $p_n\# \pm 1$  prime infinitely often.
- 24. New Mersenne Conjecture:** For odd prime  $p$ , conditions  $p = 2^k \pm 1$  or  $4^k \pm 3$ ,  $2^p - 1$  prime,  $(2^p + 1)/3$  prime are equivalent in stated pairing.
- 25. Sierpinski minimal:** 78557 is smallest  $k$  with  $k2^n + 1$  composite for all  $n$ .
- 26. Riesel minimal:** 509203 is smallest  $k$  with  $k2^n - 1$  composite for all  $n$ .
- 27. Euclid-Mullin contains all primes:**  $a_1 = 2$ ,  $a_{n+1} =$  smallest prime factor of  $1 + \prod_{i \leq n} a_i$ . Does  $\{a_n\}$  contain every prime?
- 28. Brocard's Problem:**  $n! + 1 = m^2$  only  $n = 4, 5, 7$ .
- 29. Goldbach Strong:** Every even  $n > 2 = p + q$ ,  $p, q$  prime.
- 30. Polignac's Conjecture:** For every even  $k$ , infinitely many  $p$  with  $p + k$  prime.
- 31. Sophie Germain infinitude:** Infinitely many  $p$  with  $2p + 1$  prime.
- 32. Landau  $n^2 + 1$ :** Infinitely many primes  $n^2 + 1$ .
- 33. Prime triplet infinitude:** Infinitely many  $(p, p + 2, p + 6)$  and  $(p, p + 4, p + 6)$ .
- 34. Prime quadruplet infinitude:** Infinitely many  $(p, p + 2, p + 6, p + 8)$ .
- 35. Collatz:** For  $n > 0$ ,  $n \rightarrow n/2$  if even,  $3n + 1$  if odd, reaches 1.
- 36. Vojta's Conjecture:** For  $X$  smooth,  $D$  normal crossings,  $h_{K_X+D}(P) \leq N^{(1)}(D, P) + d(P) + O(1)$ . Implies abc, Szpiro, Pillai, Hall, Fermat-Catalan.
- 37. Beal's Conjecture:** If  $A^x + B^y = C^z$  with  $x, y, z \geq 3$ , then  $\gcd(A, B, C) > 1$ .
- 38. Erdős-Straus:** For all  $n \geq 2$ ,  $4/n = 1/x + 1/y + 1/z$  in positive integers.
- 39. Erdős-Moser equation:**  $1^k + \dots + (m-1)^k = m^k$  only  $1 + 2 = 3$ .
- 40. Perfect Cuboid:** Does box with integer edges, face diagonals, space diagonal exist?
- 41. Sums of three cubes:** For each  $k \not\equiv \pm 4 \pmod 9$ , exists  $x^3 + y^3 + z^3 = k$ .
- 42. Littlewood Conjecture:**  $\inf_{n \geq 1} n \|n\alpha\| \|n\beta\| = 0$  for any real  $\alpha, \beta$ .
- 43. Bloch-Kato Conjecture:** For motive  $M$ ,  $\text{ord}_{s=0} L(M, s)$  equals Selmer rank and special value given by regulator  $\times$  Sha etc. Implies BSD.
- 44. Standard Conjectures:** Numerical equivalence = homological equivalence for cycles, Lefschetz standard, Hodge standard. Implies Hodge and Tate.
- 45. Fontaine-Mazur:** Every irreducible odd  $p$ -adic Galois representation unramified outside finite set and de Rham at  $p$  comes from geometry.
- 46. Langlands Reciprocity:**  $n$ -dimensional Galois representations correspond to automorphic representations of  $GL(n)$ .
- 47. Artin Holomorphy:** Artin  $L$ -functions for non-trivial irreducible Galois representations entire.
- 48. Elliott-Halberstam:**  $\sum_{q \leq x^{1-\epsilon}} \max_{(a,q)=1} |\pi(x; q, a) - \pi(x)/\phi(q)| \ll x/(\log x)^A$ .
- 49. Sarnak Möbius Randomness:**  $\sum_{n \leq N} \mu(n) f(T^n x) = o(N)$  for zero-entropy  $T$ . Implies Chowla.
- 50. Sato-Tate general:** For genus  $> 1$ , Frobenius angles equidistributed via Haar measure of Sato-Tate group.
- 51. Montgomery Pair Correlation:** Rescaled zeros of  $\zeta$  have pair correlation  $1 - (\sin \pi u / \pi u)^2$ .
- 52. Keating-Snaith:**  $\frac{1}{T} \int_0^T |\zeta(1/2 + it)|^{2k} dt \sim a_k (\log T)^{k^2}$ .
- 53. Gauss Circle:**  $N(R) = \#\{(x, y) \in \mathbf{Z}^2 : x^2 + y^2 \leq R^2\} = \pi R^2 + O(R^{1/4+\epsilon})$ .
- 54. Dirichlet Divisor:**  $\sum_{n \leq x} d(n) = x \log x + (2\gamma - 1)x + O(x^{1/4+\epsilon})$ .
- 55. Piltz divisor:**  $\sum_{n \leq x} d_k(n) = x P_{k-1}(\log x) + O(x^{(k-1)/2k+\epsilon})$ .
- 56. Waring  $G(4) = 16$ :** Every large integer sum of at most 16 fourth powers.
- 57. Carmichael Totient:** No  $m$  has exactly one  $n$  with  $\phi(n) = m$ .
- 58. Scholz addition chain:**  $l(2^n - 1) \leq n - 1 + l(n)$  where  $l$  minimal addition chain length.
- 59. Leopoldt's Conjecture:**  $p$ -adic regulator  $R_p(K) \neq 0$ .
- 60. Vandiver's Conjecture:**  $p \nmid h_p^+$ ,  $p$  does not divide class number of real subfield of  $\mathbf{Q}(\zeta_p)$ .
- 61. Iwasawa  $\mu = 0$ :**  $\mu$ -invariant of cyclotomic  $\mathbf{Z}_p$ -extension is 0.
- 62. Greenberg's Conjecture:** For totally real field, Iwasawa  $\lambda = \mu = 0$  for cyclotomic  $\mathbf{Z}_p$ -extension.
- 63. Real quadratic class number 1 infinitude:** Infinitely many  $d > 0$  squarefree with  $h(\mathbf{Q}(\sqrt{d})) = 1$ .
- 64. Goormaghtigh:** Only solutions to  $(x^m - 1)/(x - 1) = (y^n - 1)/(y - 1)$ ,  $m, n > 2$ ,  $x \neq y$  give 31 and 8191.
- 65. Singmaster's Conjecture:** Multiplicity of  $n > 1$  in Pascal's triangle bounded by absolute constant, conjectured  $\leq 8$ .

66. **Erdős-Turán additive bases:** If  $A$  asymptotic basis order 2, representation function  $r_A(n)$  unbounded.
67. **Erdős AP Conjecture:** If  $A \subset \mathbb{N}$  and  $\sum_{a \in A} 1/a$  diverges,  $A$  contains arbitrarily long arithmetic progressions.
68. **Erdős-Selfridge:**  $(m+1) \cdots (m+k)$  never perfect power.
69. **Lonely Runner:**  $k$  runners with distinct speeds on unit circle, some time each distance  $\geq 1/(k+1)$  from origin.
70. **Frankl Union-Closed:** In finite union-closed family containing non-empty set, some element in at least half sets.
71. **Davenport constant:**  $D(G)$  minimal  $n$  such that any sequence of  $n$  elements of abelian group  $G$  has non-empty zero-sum subsequence. Exact values open.
72. **EGZ constant:**  $s(G)$  minimal  $\ell$  such that any sequence of  $\ell$  elements has  $|G|$  elements summing to zero. Exact values open in higher rank.
73. **Erdős-Fuchs:** If  $\sum_{k \leq n} r_A(k) = cn + o(n)$  then  $o(n)$  not  $o(n^{1/4}/\sqrt{\log n})$ .
74. **Lander-Parkin-Selfridge:** If  $\sum_{i=1}^m x_i^k = \sum_{j=1}^n y_j^k$  with  $m+n < k$ , no non-trivial solutions for  $k \geq 4$ .
75. **Schanuel's Conjecture:** If  $z_1, \dots, z_n$  linearly independent over  $\mathbb{Q}$ , then  $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(z_i, e^{z_i}) \geq n$ .
76. **Normality of  $\pi$ :** Every  $k$ -digit string appears in base 10 expansion of  $\pi$  with density  $10^{-k}$ .
77. **Borel's Conjecture:** Every irrational algebraic number normal in every base.
78. **Copeland-Erdős normal:** 0.23571113... formed by concatenating primes normal base 10.
79. **Champernowne all bases:** 0.1234567891011... normal in every base.
80. **Powers of 2 digit:** Every large  $2^n$  contains digit 7, and only finitely many  $2^n$  avoid digit 0.
81. **Prime geometric progression:** Four distinct primes in geometric progression with rational ratio exist.
82. **Gaussian Moat:** Cannot walk to infinity stepping only on Gaussian primes with bounded step.
83. **Markov unicity:** Each Markov number  $z$  appears as max of exactly one triple  $x^2 + y^2 + z^2 = 3xyz$ .
84. **Margenstern:** Every large even integer sum of two practical numbers.
85. **Odd perfect number existence:** Does  $n$  with  $\sigma(n) = 2n$  odd exist?
86. **Quasiperfect existence:** Does  $n$  with  $\sigma(n) = 2n+1$  exist?
87. **Odd weird existence:** Does odd abundant non-semiperfect exist?
88. **Amicable pairs infinitude:** Infinitely many  $a \neq b$  with  $\sigma(a) = \sigma(b) = a+b$ .
89. **Sociable chains infinitude:** For each  $k$ , infinitely many aliquot cycles length  $k$ .
90. **Catalan-Dickson:** Every aliquot sequence  $n, \sigma(n) - n, \dots$  eventually 0 or periodic.
91. **Lychrel 196:** 196 never becomes palindrome under reverse-and-add.
92. **Mahler 3/2:** Fractional parts  $\{(3/2)^n\}$  not dense,  $\inf\{(3/2)^n\} > 0$ .
93. **Primary pseudoperfect infinitude:**  $n$  with  $\sum_{p|n} 1/p + 1/n = 1$  infinitely many.
94. **Superperfect infinitude:**  $\sigma(\sigma(n)) = 2n$  infinitely many.
95. **Giuga numbers infinitude:** Composite squarefree  $n$  with  $p | n \Rightarrow p | n/p - 1$  infinitely many.
96. **Lucas-Carmichael infinitude:** Composite squarefree  $n$  with  $p | n \Rightarrow p+1 | n+1$  infinitely many.
97. **Primitive non-deficient infinitude:** Minimal abundant numbers w.r.t divisibility infinitely many.
98. **Primitive weird infinitude:** Minimal weird numbers infinitely many.
99. **Hilbert Tenth over  $\mathbb{Q}$ :** No algorithm decides existence of rational solution to Diophantine equation.
100. **Unbounded ranks:** Elliptic curves over  $\mathbb{Q}$  have sup rank  $= \infty$ .
101. **Colliot-Thélène:** Brauer-Manin obstruction only obstruction to Hasse principle for rationally connected varieties.
102. **Uniformity of rational points:** Number rational points on genus  $g \geq 2$  curve over  $\mathbb{Q}$  bounded by  $C(g)$ .
103. **Bombieri-Lang:** Rational points on variety of general type over number field not Zariski dense.
104. **Cohen-Lenstra heuristics:** Distribution of class groups of quadratic fields follows predicted probabilities.
105. **Stark Conjectures:** Leading term of Artin  $L$  at  $s=0$  equals regulator of Stark units.
106. **Generalized Ramanujan-Petersson:** Hecke eigenvalues of cuspidal automorphic forms on  $GL(n)$  have absolute value 1.
107. **Irrationality of  $\gamma$ :**  $\gamma = \lim(H_n - \log n)$  irrational.
108. **Irrationality of  $\zeta(2k+1)$ :**  $\zeta(5), \zeta(7), \dots$  irrational.
109. **Algebraic independence odd zeta:**  $\zeta(3), \zeta(5), \dots$  algebraically independent over  $\mathbb{Q}$ .

- 110. Catalan's constant irrational:**  $G = \sum_{n \geq 0} (-1)^n / (2n+1)^2$  irrational. with  $p, q$  prime.
- 111. Erdős primitive sets:** For primitive  $A$  (no element divides another),  $\sum_{a \in A} 1/(a \log a) \leq \sum_p 1/(p \log p)$ .
- 112. Normality of  $e$ :**  $e$  normal base 10.
- 113. Lemoine's Conjecture:** Every odd  $n > 5 = p + 2q$
- 114. Agoh-Giuga:**  $n$  prime iff  $B_{n-1} \equiv -1 \pmod n$  and  $\sum_{p|n} 1/p - \prod_{p|n} 1/p \in \mathbf{N}$ ,  $B$  Bernoulli.
- 115. Erdős-Ulam:** Does dense set in plane with all mutual distances rational exist?